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ARRAY QUESTIONS (10)

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1. Write a program to swap the first and last elements of an array
WITHOUT using any extra array or temporary variable.
2. Given an array of integers, count how many elements are prime numbers.
Do NOT use any predefined prime-checking function.
3. Write a program to reverse an array IN PLACE
WITHOUT using another array.
4. Find the second largest element in an array
WITHOUT sorting the array.
5. Write a program to check whether an array is sorted in ascending order
WITHOUT using any sorting function.
6. Given an array, replace every element with the square of that element.
If the square overflows int range, print "Overflow".
7. Write a program to rotate an array to the right by one position
WITHOUT using extra memory.
8. Find the frequency of a given element in an array
WITHOUT using any additional array.
9. Write a program to find the smallest and largest elements in an array
using only ONE loop.
10. Given an array, print all elements that appear only once.
Do NOT use maps, sets, or additional arrays.

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STRING QUESTIONS (10)

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1. Reverse a string WITHOUT using any built-in reverse function
and WITHOUT using an extra string.
2. Write a program to check whether a string is a palindrome
WITHOUT using string comparison functions.

3. Count the number of vowels, consonants, digits, and spaces in a string using a single loop.
4. Convert all lowercase letters in a string to uppercase WITHOUT using built-in case conversion functions.
5. Write a program to find the length of a string WITHOUT using `strlen()` or `string.length()`.
6. Remove all duplicate characters from a string WITHOUT using any extra data structure.
7. Count how many words are present in a string WITHOUT using `stringstream`.
8. Write a program to toggle the case of every alphabet in a string (uppercase $\rightarrow$ lowercase, lowercase $\rightarrow$ uppercase).
9. Check whether a string contains only digits.
If not, print the first non-digit character found.
10. Write a program to replace all spaces in a string with '_' WITHOUT using `replace()`;
11. Write a program to convert decimal numbers to binary number.
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LOGIC-BASED (BASICS ONLY)
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1. Write a program to swap two numbers WITHOUT using a third variable and WITHOUT using arithmetic operators.
2. Write a program to calculate factorial of a number using ONLY a while loop.
Reject negative input.
3. Write a program to calculate power (a^b) WITHOUT using `pow()` and WITHOUT recursion.
4. Write a program to check whether a number is prime by minimizing the number of divisions.

5. Write a program to reverse a number.
6. Write a program to check whether a number is an Armstrong number
WITHOUT converting the number to a string.
7. Write a program to count the number of digits in a number
WITHOUT using log functions.
8. Write a program to find the sum of digits of a number
until a single-digit result is obtained.
9. Write a program to print all prime numbers between 1 and n
WITHOUT using arrays.
10. Write a program to check whether a number is a perfect square
WITHOUT using sqrt().